FH University of Applied Sciences TECHNIKUM WIEN

Master Thesis Poposal

In AI Engineering

Enhancing financial forecasting by sentiment analysis

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1 Abstract

Financial markets are highly complex systems that are influenced by human behaviours and economic events. For decades, data scientists and economists have attempted to predict and explain financial time series, but it still remains a big challenge. Traditionally, autoregressive models or technical indicators were used to achieve this goal. While this approach created a mathematical and statistical foundation, it falls short when confronted with unstructured data like news headlines or investor sentiment.

With the rapid advances in artificial intelligence (AI), especially in natural language processing (NLP), new ways for understanding those unstructured texts have been found. For example, transformer models can extract semantic information from text. This information could be used to engineer new features that help explain variance in the movement of stock market indices like the S&P500 or the MSCI World.

The goal of this master thesis, therefore, is to investigate if a stochastic forecasting approach can be improved by adding sentiment-based features extracted from news headlines and/or social media posts. This feature can be understood as an indicator (usually ranged between -1 and 1), that tries to reflect the emotions of investors towards the current economy. In the scope of a single headline it captures how optimistic (bullish) or pessimistic (bearish) the context appears.

As a proof of concept (PoC), the work will analyse an ARMA model with and without sentiment integration. Later, more complex AI models – ranging from XGBoost to recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and state-of-the-art transformer models like GPT – 5 may be evaluated and benchmarked. To analyse performance, metrics such as Mean Percentage Error (MPE), Accuracy, and F1-Score will be used.

This investigation contributes to an ongoing debate about the efficient market hypothesis, which states that new information is integrated into prices almost instantly, rendering the information useless for predicting future movements.

2 Motivation

Despite decades-long research in terms of stock predictions, the forecasting of short-term movements remains a challenge. Traditional methods depend solely on previous price movements and technical indicators. However, the influential factors that move the market go far beyond this information. News, Social Media, politics, macro-economic events are examples that could contribute to changes in the stock market as well.

Current studies [1] increasingly work with the analysis of media texts to achieve additional prediction-accuracy, though, the quantitative value of these features is debated. Additionally, the interaction between technical indicators and sentiment features — and their impact on model performance across different AI architectures — remains an open question.

3 Literature Review

The idea of combining market sentiment with stochastic and technical indicators is not new. Many studies have already done research on it [1] [2] [3]. The combination of sentiment and well known indicators like ADI (divergences between the price and the volume), volume, EMA (weighted moving average) are used in [1] to predict the price changes in Forex (Foreign exchange markets), with good performance. Similarly [3] tried using daily aggregated sentiment and stock-data (isn´t further explained) to predict price changes of certain companies like Apple, Amazon or Microsoft. Both works have proven a correlation between stock price and sentiment, which led to increased model performance. Also notable is that [1] stated that specifically combining both technical indicators and sentiment led to the most performant model, hinting possible feature synergies or complementations.

However, newspaper headlines are not the only reliable source, social media posts (e.g. twitter or social hubs) can also be utilized to create sentiment features. For example [4] used properties like views, likes, dislikes, and keywords to predict price momentum (up, down). Other studies like [5] even go a step further by combining both news paper headlines with social media content to create more robust accuracy.

In comparison to others this study will focus not on single companies or foreign exchange markets but broad stock indices. One might argue that sentiment analysis is more effective for indices because it reflects more macroeconomic factors. However, the opposite could also be true. Headlines related to a specific company (not included by the index) may introduce

noise into the aggregated sentiment score, potentially reducing predictive power. Comparing the outcome of this study with the related ones may give new insights on this topic.

4 Proposed Methods

The fundamental data of this work will be a big corpus of news paper headlines, that combine multiple American media sources (CNBC, Reuters, The Guradians, ...). A pipeline that combines many of those datasets will be implemented. Primary goal ist to unify many different datasets, remove duplicates and create a time series with news information. On the other hand using investpy or other public APIs (Alpha Vantage) stock and volume data will be collected for the target index. The frequency will be adjusted for the granuality of the sentiment data (for example an intra day sentiment score is only possible if enough data can be collected).

4.1 Example Features Matrix

Sentiment scores will be genreated with different algorithms and models to compare the effect on performance and stability. Starting from "primitive" tools like lexicon based analysis to state of the art fine tuned transformer models like FinBERT or LLMs like GPT-5. Technical indicators will be calculated from the past price and volume data as part of the feature matrix pipeline. A possible matrix could look like this:

Table 1: Possible Feature Matrix

Feature	Description
Close Price / Price Change	The price of a fincial instrument at the end of each trading day / The price change that measures the relative or absolute change to prior days
Volume	Trading volume shows how many shares have been traded in a give time interval
Exponential Moving Average (EMA)	Moving Average that assignes greater weight to newer data points
Average True Range (ATR)	Measures volatility over a specific time period.
Williams %R	A momentum indicator that indicates whether a stock is overbought (around 0) or oversold (around -100).
Accumulation/Distribution Indicator (ADI)	An indicator that combines price and volume data to determine the capital flow into a stock.
Sentiment Score	daily aggregated score, that tries to predict market sentiment

(Disclaimer: The provided feature matrix is only an idea. Both technical indicators and sentiment features can/will be changed)

4.2 Example Metrics and Statistical Tests

For regression forecasting tasks (price changes), the following metrics can be used: **Mean Percentage Error (MPE)**, **Mean Absolute Error (MAE)**

When predicting directional movements (up, down), the following classification metrics can be applied: **Accuracy**, **Precision**, **Recall**, **F1-Score**

To validate whether sentiment features have a statistically significant impact on model performance, the following tests could be used: **p-Value Test**, **Granger Causality Test**

4.3 Goal and Milestones

The goal of this study is to quantify the benefit of market-sentiment to predict changes in stock prices. Different AI-algorithms will be compared and analysed to determine which would benefit the most from this new feature. Following milestone list will outline the work:

Table 2: Goals Milestone Plan

Week / Month	**Task**	**Status**
Week 1–2 (Sept)	Start with Literature Review and scouting for the first prototyp datasets of news headlines	Done
Week 3–4 (Sept)	Create a simple sentiment pipeline that classifies news headlines. Analyse different finance APIs that provide daily prices for finacial products	Done
Week 5–6 (Oct)	Use the gathered and prepared data to train ARMA Models and compare the effect of adding the sentiment feature. (first prototype)	Done
Week 7–10 (Oct/Nov)	Create a pipeline that can combine different dataset, gather more data and try around with different sentiment lags (analyse the decay of explainability over time) / compare sentiment models (lexicon -> transformer)	Pending
Week 11–12 (Nov)	Create pipeline that calculates technical indicators and combine them with the sentiment scores to create a complete feature Matrix	Pending
Week 13–16 (Dec)	Model training and evaluation (ARMA, XGBoost, RNN, LSTM, GPT) Evaluation metrics (MPE, Accuracy, F1-Score)	Pending
Week 17–18 (Jan)	feature contirbution analysis, performance comparision and preparation of presentation	Pending
Week 19–20 (Jan)	buffer	-

5 First prototype

In the course of a minimal experiment, ARIMA-Models produced hints that sentiment has a statistical value for predicitng prices. For this purpose, an autoregressive (AR) model with a moving average (MA) component was used. ARIMA models are primary used in statistic and economic szenarios. It is important to note that, in this case the previous proposed matrix was not used. Instead only price and sentiment of prior N days where utilized for predicting future values.

A ARIMA(p,d,q) model has 3 different parameter [6]:

- p => N past datapoints (t-1, t-2, ... t-n)
- d => degree of differencing, makes changes stationary (from price movement to price change)
- q => N past regression errors

5.1 Setup

Table 3: Setup of the experiment

Model	Arima(1,0,1) vs ARIMA (1,0,1) + daily aggregated sentiment
Dataset	- Kaggle dataset (CNBC, Retuers, The Guardians - headlines) covering 01.01.2018 - 16.07.2020 [7] - daily prices of the S&P500 (US) in the same time range via investpy API (648 data points)
Sentiment Model	Vader Lexikon
Benchmark	AIC, BIC

5.2 Results

```
Best ARMA(1,1) AIC: -3633.634057451844, BIC: -3615.7384946664347
Best ARMA with sentiment (1,1) AIC: -3638.885825629695, BIC: -3616.5163721479335
```

Figure 1: AIC and BIC Benchmark

Both AIC and BIC shrink with the addition of sentiment score. Even though the difference is small, this experiment provide valuable insights that sentiment can contribute positive to model performance.

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0002	0.001	0.374	0.709	-0.001	0.001
daily_sentiment	0.0199	0.008	2.594	0.009	0.005	0.035
ar.L1	-0.5013	0.045	-11.184	0.000	-0.589	-0.413
ma.L1	0.2269	0.051	4.491	0.000	0.128	0.326
sigma2	0.0002	4.74e-06	44.240	0.000	0.000	0.000

Figure 2: Feature list ARMA(1,0,1) with sentiment

The low p-value proves that the feature daily_sentiment is statistical significant

While this results are very promising following two challenges were indentified. First, while the ARMA sentiment model generally outperformed the ARMA mdoel, this was not the case for every configuration. Second for the year 2018, no significant relationship between sentiment and price movement could be proven (p-value $\bar{0}$.457). Therefore, although this experiment is an important indicator that sentiment explains some variance, more data and further investigations are needed to find clear answers.

6 Example Plots

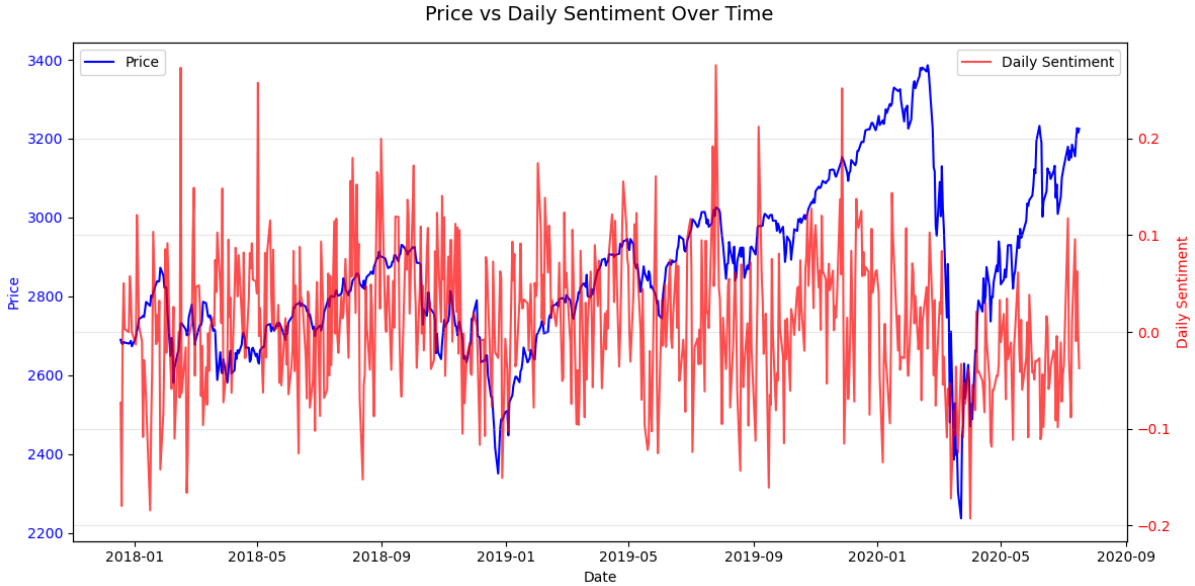


Figure 3: Sentiment vs Price

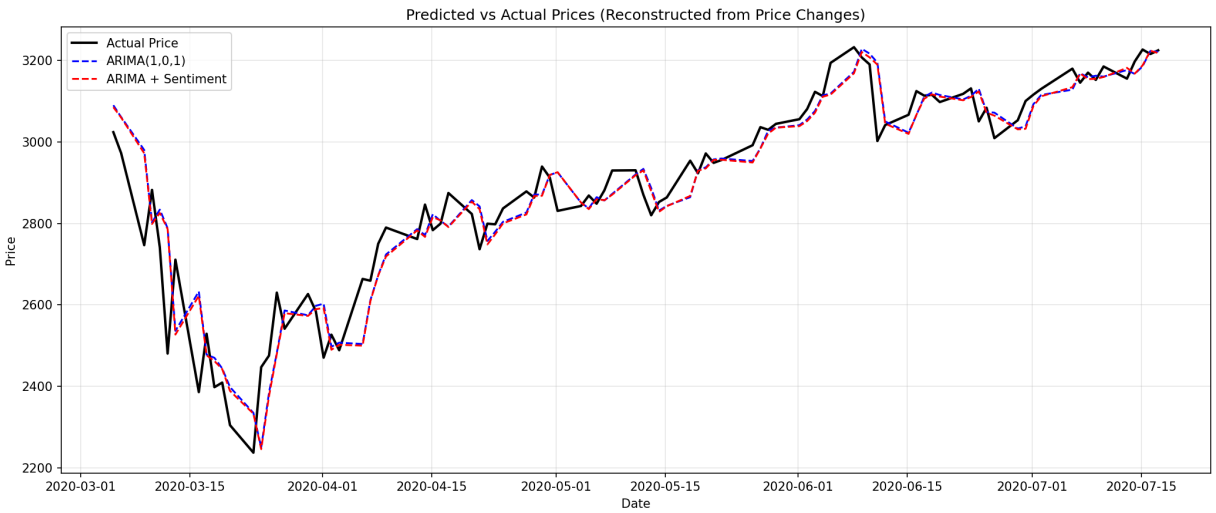


Figure 4: Predicted vs Actual Prices

7 Progress Report

7.1 Lookahead BIAS

After the ARMA experiment, the next step was to use more capable sentiment models than the vader lexicon. The expected outcome with FinBERT was a better performance, because the representation of the sentiment score is closer to the ground truth. The results were suprisingly good, too good in fact. This lead to some scepsis about the results. After investigating the codebase, a data leak was indentified in the training process of the ARMA model. The model had access to the headlines of tomorrow which lead to an unfair edge. While this finding changed the optimism of the use of a sentiment feature as a usefull predictor, it provided evidence that a better sentiment model leads to a better representation of the market sentiment. The positive correlation of tomorrow sentiment and tomorrow price change increased which means that the FINBERT model captchered better and more usefull information of the news headlines. (Positive correlation went from 0.0220 to 0.0283, with the AIC score decreasing by additional 21.96 points)

7.2 Feature Matrix Pipeline

To validate the results of the experiment across multiple datasets and sentiment models, a pipeline for the feature matrix was build. Following figure shows the top level architecture.

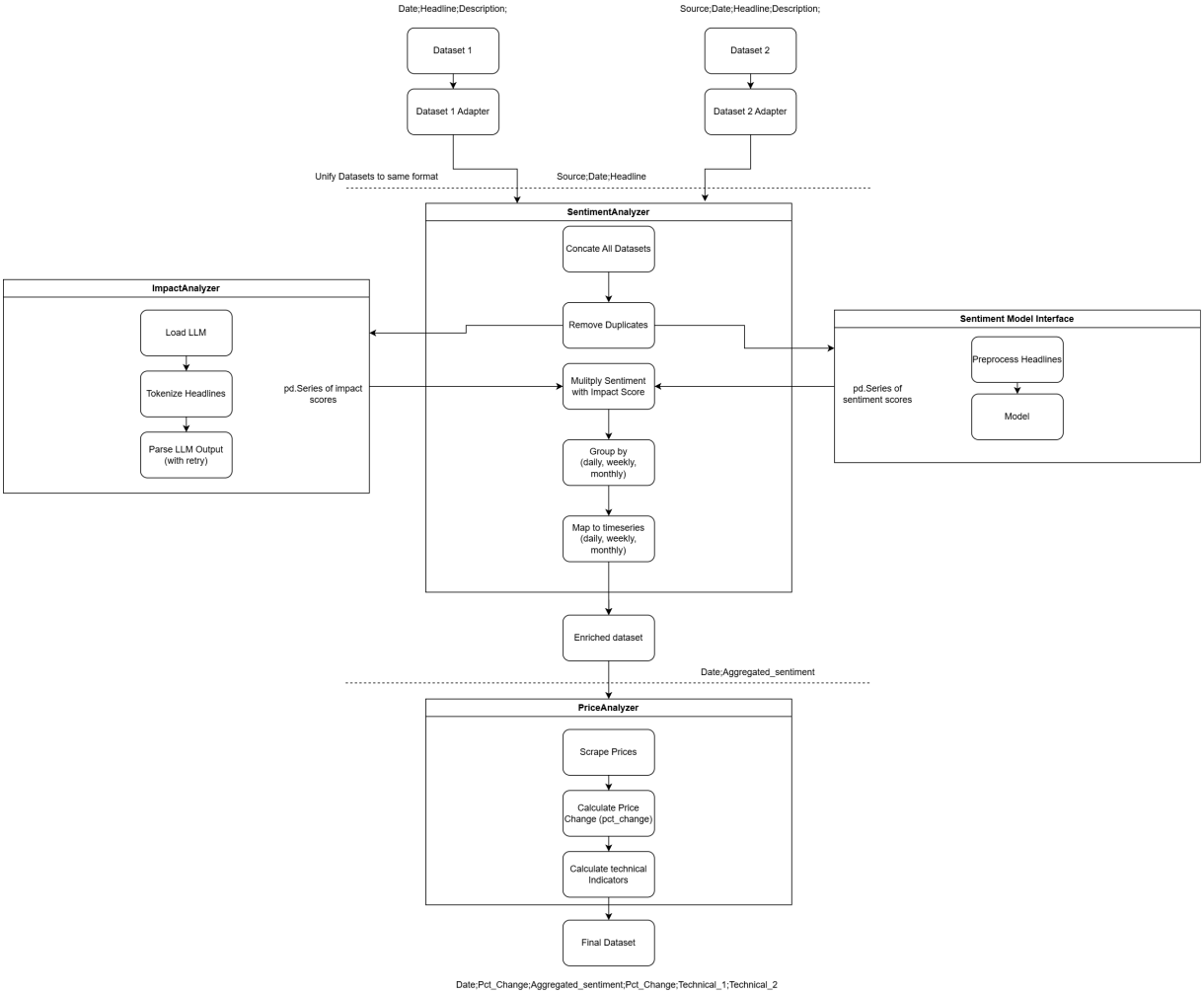


Figure 5: Feature Matrix Pipeline - Architecture

7.2.1 Sentiment Analyzer

This part of the pipeline is used to take an arbitrary number of financial news datasets unify, concatenate and analyze them and output an aggregated sentiment score for each date. The modular design allows to perform experiments on a variety of different datasets and sentiment models. Resulting in an efficient way to test this works hypothesis on sentiment scores.

To achieve the concatenation part of the pipeline. Each dataset has its own adapter, for transform rows in a standard format. The standard format looks like this: source;date;headline. After deduplication a VADER, FinBERT or similar models are used to create the sentiment score. And lastly those scores are grouped by granuality (week, day) and mapped to a timeseries.

7.2.2 Impact Score Analyzer

Many available (financial) news headline datasets have a lot of noise. For example, headlines about crimes, financial quizzes or news about foreign economics do not have the same importance than fed reserve news or US politics. Therefore in this work a novel way of denoising is introduced. An LLM is prompted to extract the importance of a given headline for the s&p500 next day price change prediction. Similar to the sentiment analyzer the output is a panda series of impact scores, ranging von 0 (not important) to 1 (very important). After extracting the importance of a headline. The impact score is then multiplyed with the sentiment score resulting in denoised weighted sentiment score.

7.2.3 Price & Technical Indicators

For the technical indicators and price change data, this work used investpy. Specifically, the investpy.etfs.get_etf_historical_data endpoint. The minimal granuality of investpy is daily. Because of this limitation the minimal granuality of the final feature matrix cannot be intra day (hourly). After scraping the closing prices of 'SPDR S&P 500', price change data and technical indicators like (Volume, volatility) are calculated and added to the final feature matrix.

Following API is used as a starting point for each experiment:

```
start_date = "17/12/2017"
end_date = "06/11/2020"
impact_model = ImpactModel.NONE
df_combined = get_feature_matrix(
    start_date=start_date,
    end_date=end_date,
    impact_model=impact_model,
    tech_indicators=[TechnicalIndicators.VOLATILITY],
    sentiment_sources=[DatasetSources.FNSPID],
    sentiment_model=SentimentModel.FINBERT,
    granularity_level=GranularityLevel.DAILY
)
```

Figure 6: Feature Matrix API

7.3 Findings

7.3.1 Negative Correlation

Across the datasets that has been used in the experiments (NIFTY, FNSPID, LucasPhamFin-Dataset). A similar trend has been indentified. Between sentiment score of today and price change of tomorrow is a **negativ correlation**. This correlation was measured via Pearson and Spearman correlation coefficient. This hints that the efficient market hypothesis holds for the

'SPDR S&P 500' etf. Information of headlines are included in the price within <24 hours, making future predictions less effective or useless. One possible explaination could be that investors already taking profits of the good news or using a "buy the dip" strategy for when the sentiment is negative.

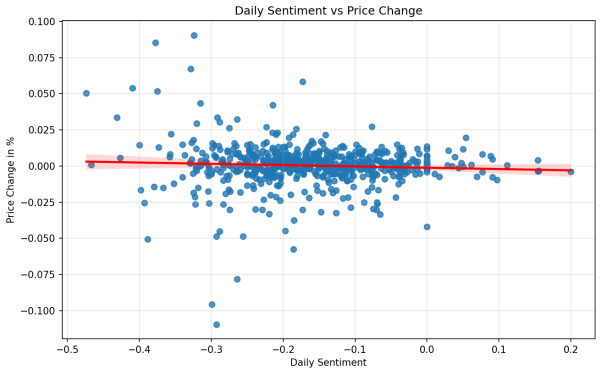


Figure 7: Point cloud of sentiment and price change

7.3.2 Sentiment and volatility

Another experiment compared the aggregated sentiment and the volatility (standard deviation of the last 30d log price change). Here a clear negative correlation could be indentified. If market sentiment is bad, volatility rises. If sentiment is high volatility stabilizes. This is an intersting finding hinting that market mood is linked to price volatility.

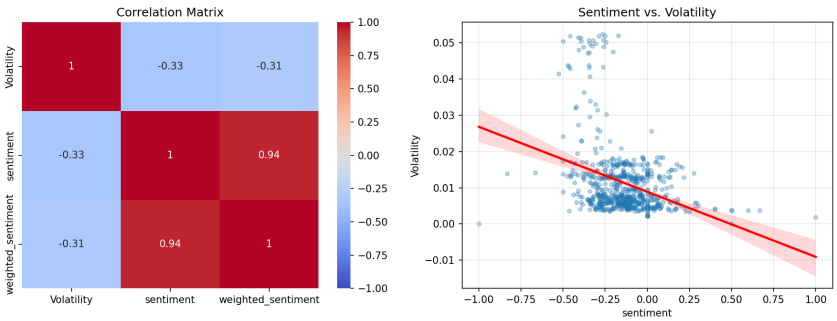
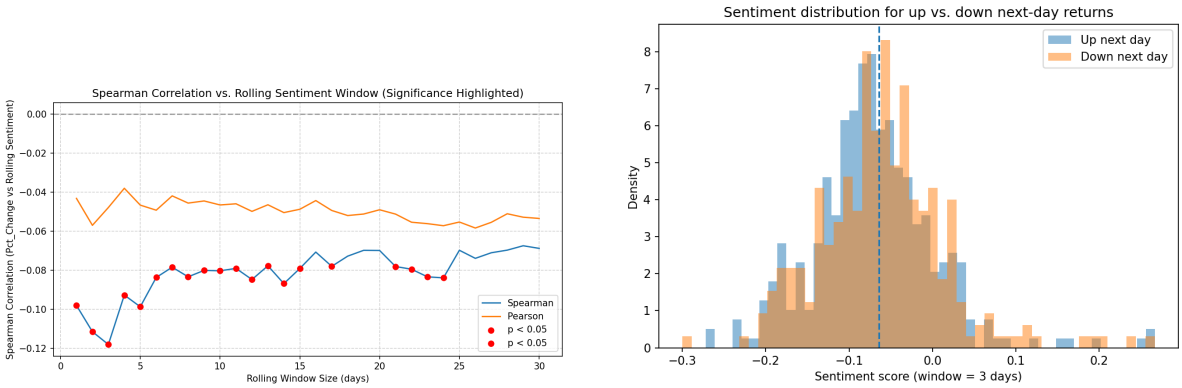


Figure 8: Linear Sentiment x Volatility correlation

Notable a linear function seem to not fit the data well, because volatility cannot be negative and also it seems like there is a certain threshold rule to be found here.

7.3.3 Rolling sentiment

Similar to the experiment with sentiment and price change the rolling sentiment compares the last 1-30 days rolling sentiment to the next day price change. The goal was to indentify a rolling window that has the strongest spearman correlation and create two histograms of sentiment and price change. If the two histograms can be seperated by a line on the x-axis we could use the value as thershold rule. Here the negative correlation can be found again, indicating that longer time horizons of sentiment does not help overcome the efficient market limitations.



(a) Pearson & Spearman correlation

(b) Histogram seperation

Figure 9: Rolling weighted sentiment and next day price change

As seen in 9b the seperation of the two histograms is not optimal. As stated by the correlation the histograms also indicate that positive sentiment values lead to negative returns on the next day. Important to note is that adding the impact score did improve the seperation by a small amount (Youden´s J improved from -0.0743 to -0.1068, where 1 or -1 are perfect splits)

7.3.4 Better performance with trend prediction

The experiments with sentiment and weighted sentiment showed a overall negative correlation between the predictor and tomorrow price change. The next approach was to predict a longer time horizon, giving the price returns time to balance out. The idea is that predicting trends would yield more stable and better performance. The new target variables are 3d, 5d and 20d returns.

The results showed that accumulating future returns yield better performance. Especially, the 3d and 5d target gave the ARIMA model with sentiment a greater edge compared to the experiment with 1d returns.

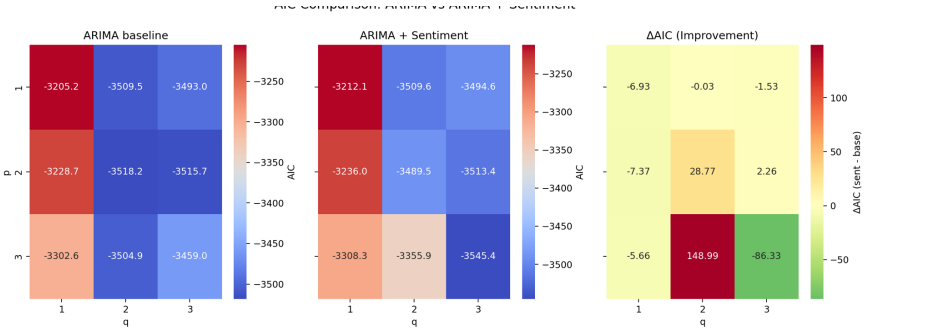


Figure 10: 3d Target ARMA Results (Best AIC improved by 27 points)

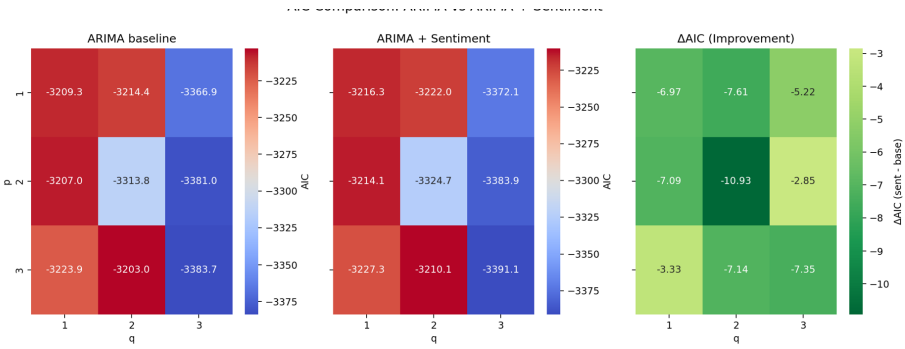


Figure 11: 5d Target ARMA Results (Overall Improvement of the AIC)

The 20d target worked worse than the 1d Target, indicating that today sentiment is not important to predict long time horizons like 20d. If the new targets will also improve non linear models like XGBoost is unknown at this point, and is another experiment in the future.

7.3.5 Non Linear Models

To capture non-linear relationships, an experiment with XGBoost is tasked to classifying the next day return as up or down. Here an improvement in both recall and precision can be achieved by adding weighted sentiment to the model compared to a decision tree with only price change as feature(F1-score 0.5513 improved to 0.6380). If Volatility and sentiment are combined the F1-Score further improves. Howerver the magnitude of this improvement is dependent on the dataset. For example using only LucasPhamFinDataset (like in the other experiments) the increase in model performance is neglegtable (0.0037). If combined with additional headlines from the NIFTY Dataset the improvement is significant (0.0322) which indicate that the quality and amount of headlines matter with the interaction of volatility and weighted sentiment.

Following metrics where measured (LucasPhamFinDataset only):

Features Used	Accuracy	Precision	Recall	F1-score
Pct_Change	0.4615	0.5244	0.5811	0.5513
Pct_Change + Weighted_Sentiment	**0.5462**	**0.5843**	0.7027	0.6380
Pct_Change + Volatility	0.5308	0.5607	0.8108	0.6630
Pct_Change + Weighted_Sentiment + Volatility	0.5385	0.5660	**0.8108**	**0.6667**
Pct_Change + Volatility + Volume	0.5231	0.5577	0.7838	0.6517
Pct_Change + Weighted_Sentiment + Volatility + Volume	0.5077	0.5500	0.7432	0.6322

Figure 12: Results of XGBoost on LucasPhamFinDataset

Overall the expermient with XGBoost showed a consistent improvement when adding weighted sentiment to the feature list. A non linear model seems to learn something from the negative correlation.

7.4 Milestones and next steps

Table 4: Goals Milestone Plan

Week / Month	**Task**	**Status**
Week 1–2 (Sept)	Start with Literature Review and scouting for the first prototyp datasets of news headlines	Done
Week 3–4 (Sept)	Create a simple sentiment pipeline that classifies news headlines. Analyse different finance APIs that provide daily prices for finacial products	Done
Week 5–6 (Oct)	Use the gathered and prepared data to train ARMA Models and compare the effect of adding the sentiment feature. (first prototype)	Done
Week 7–10 (Oct/Nov)	Create a pipeline that can combine different dataset, gather more data and try around with different sentiment lags (analyse the decay of explainability over time) / compare sentiment models (lexicon -> transformer)	**Done**
Week 11–12 (Nov)	Create pipeline that calculates technical indicators and combine them with the sentiment scores to create a complete feature Matrix	**Done**
Week 13–16 (Dec)	Model training and evaluation (ARMA, XGBoost, RNN, LSTM, GPT) Evaluation metrics (MPE, Accuracy, F1-Score)	Pending
Week 17–18 (Jan)	feature contirbution analysis, performance comparision and preparation of presentation	Pending
Week 19–20 (Jan)	buffer	-

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